

Kingsley Fong

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Education

University of Waterloo | Mechanical Engineering
Honors & Co-op | President's Entrance Scholarship

2024 – 2029
Waterloo, ON

Experience

Manufacturing Engineering Intern

Jan. 2026 – Apr. 2026

S&C Electric

Etobicoke, ON

- Engineered 2 line-side **manufacturing fixtures** for packaging & **assembly operations** reducing cycle time by **~30%** with **SolidWorks, BOMs, sheet-metal drawings and machine shop/vendor coordination**.
- Built **Power BI manufacturing review dashboards** from **Azure-linked TeamAssurance data tables**, reducing weekly safety/hazard review time by **~64%** through **KPI views, drilldowns, and escalation tracking**.
- Developed **hardware material-flow tracker** with Excel VBA workflows, reorder triggers, shortage logic, and inventory health status, reducing part lookup and inventory review time by **~70%**.
- Drove **continuous improvement studies** on constrained **assembly lines**, identifying **bottlenecks** and non-value-added handling to support **+8% throughput** and **~12% cycle-time reduction**.
- Implemented **barcode Kanban system** for line-side hardware control, using Excel VBA-generated cards and bin validation to reduce replenishment errors and material downtime by **~25%**.

Mechanical Engineer Intern

May 2025 – Aug. 2025

Paragon Systems

Concord, ON

- Executed high-cycle **vibration, shock, tensile, environmental and IP validation** per **IEC 60529 / ISO 20653**, supporting **first-pass qualification** and traceable **compliance documentation**.
- Designed **modular test fixtures** in **SolidWorks with GD&T drawings & BOMs**, improving alignment repeatability and increasing test throughput by **~20%**.
- Standardized test procedures, calibration logs, and data-capture workflows**, reducing setup variability and **retest incidence ~20%**.
- Led centralized dust-collection upgrade from **3 to 8 stations**, increasing airflow **+10.5%** and reducing particulate-related reruns by **~22%**.
- Automated test-log post-processing using **Python-based log parsers**, extracting **KPIs and pass/fail metrics** to reduce reporting cycle time from **2h to 15min** and accelerate certification turnaround.

Product Design & Manufacturing Lead

Mar. 2025 – Dec. 2025

Guerrilla Gear

Remote (Austin, TX)

- Led **DFM/DFAM** for production release of modular weight-stack attachment, defining **tolerance stackups, material choices, and magnet retention features** for scalable manufacturing.
- Sourced regional machine shops & additive vendors reducing per-unit cost by **~13%** through geometry simplification and supplier consolidation.
- Transitioned V1 **machined steel/aluminum assembly** to V2 **thermoplastic variant**, enabling scalable on-demand fulfillment and supporting **six-figure revenue** growth.

Projects

Brick It – Autonomous LEGO® 3D Printer | SOLIDWORKS, RobotC, GD&T, DFM, Full-stack **Jan 2025 – April 2025**

- Led **mechanical architecture and full-stack web slicer development**, integrating CAD-to-toolpath logic and automated EV3 export; achieved **9.5 s/brick** cycle time and **92% placement accuracy** over 100+ builds.
- Designed and validated modular gantry, magazine, and structural components with **GD&T and FEA**; reduced part count **30%**, cut changeover **25%**, and improved end-effector stiffness **15%** for stable high-speed operation.

Trace It – Interactive Targeting System | SOLIDWORKS, Arduino, C++, CNC, FEA

Sep 2024 – Dec 2024

- Engineered belt-driven motion platform with closed-loop stepper control and IR sensing, achieving **95% detection accuracy** and stable tracking at **8 cm/s** under variable lighting.
- Applied **DFM, tolerance analysis, and rapid prototyping** (CNC + additive) to iterate structural components, improving reliability and **simplifying assembly** for repeatable builds.

Skills

CAD & Software: SOLIDWORKS (CSWA), ANSYS, GD&T, Drawings/BOMs, Power BI, SQL, Excel VBA

Manufacturing & Operations: Lean/5S, Value Stream Mapping, DFMA, Kanban Systems, Waste/cost Reduction

Design & Prototyping: CNC Machining, Additive Manufacturing, Laser Cutting, Fixture Design, Tolerance Stackups

Programming & Data: Python, C++, SQL, Excel VBA, NumPy, Pandas, MATLAB, LaTeX